

Modal Logic: Tableaux Examples

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Example 1:

$$\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \text{ in } K$$

Example 1: $\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ in K

1. $\models 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ Assumption

Start by assuming it's false.

1. $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond(A \wedge B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond A \wedge \Diamond B \rightarrow \mathbb{F} 1$

False conditional means true antecedent and false consequent.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond(A \wedge B)$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond A \wedge \Diamond B$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A \wedge B$ | $\Diamond \mathbb{T} 2$ |

True $\Diamond$ introduces world 1.1.

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B \checkmark$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge \mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 4$

True conjunction gives us both A and B at 1.1.

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \checkmark$	Assumption
2.	$\top 1, \Diamond(A \wedge B) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\top 1.1, A \wedge B \checkmark$	$\Diamond \top 2$
5.	$\top 1.1, A$	$\wedge \top 4$
6.	$\top 1.1, B$	$\wedge \top 4$
$\swarrow \qquad \searrow$		
7.	$\mathbb{F} 1, \Diamond A \checkmark \quad \mathbb{F} 1, \Diamond B \checkmark$	$\wedge \mathbb{F} 3$
8.	$\mathbb{F} 1.1, A$	$\Diamond \mathbb{F} 7$
9.	$\otimes \qquad \mathbb{F} 1.2, B$	$\Diamond \mathbb{F} 7$
	$\otimes$	

When we introduce worlds that make the $\Diamond$ sentences true, both branches close. So this is valid in **K**.

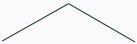
Example 2:

$$(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B) \text{ in } \mathbf{K}$$

Example 2: $(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ in $\mathbf{K}$

Let's show this is a theorem in $\mathbf{K}$.

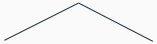
1. $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond A \vee \Diamond B \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond(A \vee B) \rightarrow \mathbb{F} 1$

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond(A \vee B)$ | $\rightarrow \mathbb{F} 1$ |
| |  | |
| 4. | $\mathbb{T} 1, \Diamond A \quad \mathbb{T} 1, \Diamond B$ | $\vee \mathbb{T} 2$ |

The disjunction branches. We can't do anything with the other line yet because there isn't yet a 1.1.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\mathbb{T} 1, \Diamond A$ ✓ $\mathbb{T} 1, \Diamond B$	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$

On the left branch, the True $\Diamond$ rule introduces world 1.1 with A true there.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{T} 1, \Diamond A \checkmark$	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
6.	$\mathbb{F} 1.1, A \vee B$	$\Diamond \mathbb{F} 3$

Now world 1.1 exists, so the False $\Diamond$ rule (line 3) applies: $A \vee B$ must be false at 1.1.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
	⊗	

The False $\vee$ rule gives $\mathbb{F} 1.1:A$, which clashes with line 5. The left branch closes. Now work on the right branch.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$ ✓ $\mathbb{T} 1, \Diamond B$ ✓	$\vee \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	⊗ $\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 4$

On the right branch, the True $\Diamond$ rule introduces world 1.2 with B true there.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$ ✓	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	$\otimes$ $\top 1.2, B$	$\Diamond \top 4$
9.	$\mathbb{F} 1.2, A \vee B$	$\Diamond \mathbb{F} 3$

World 1.2 now exists, so the False $\Diamond$ rule (line 3) applies again: $A \vee B$ must be false at 1.2.

1.	$\models 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ ✓	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$ ✓	$\rightarrow \models 1$
3.	$\models 1, \Diamond(A \vee B)$	$\rightarrow \models 1$
4.	$\top 1, \Diamond A$ ✓ $\top 1, \Diamond B$ ✓	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\models 1.1, A \vee B$ ✓	$\Diamond \models 3$
7.	$\models 1.1, A$	$\vee \models 6$
8.	⊗ $\top 1.2, B$	$\Diamond \top 4$
9.	$\models 1.2, A \vee B$ ✓	$\Diamond \models 3$
10.	$\models 1.2, A$	$\vee \models 9$
11.	$\models 1.2, B$	$\vee \models 9$
	⊗	

Both branches close, so this **is** valid in K.

Example 3: $A \rightarrow \Box \Diamond A$ in KT

Example 3: $A \rightarrow \Box\Diamond A$ in KT

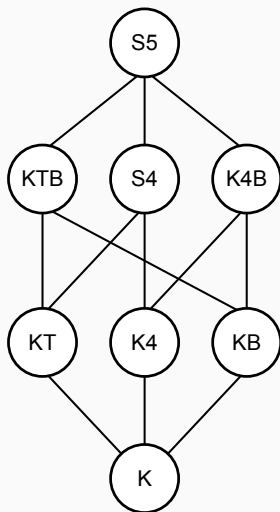
This should fail in KT. Let's show it.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, A \rightarrow \Box \Diamond A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{F} 1.1, \Diamond A$ | $\Box \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1.1, A$ | $\mathbb{T} \Diamond 4$ |

The tree stays open. So this is not valid in **KT**.

Note that after line 4 we've completed all the rules we needed to use in **K**, so the first four lines show it is not a theorem in **K**.

The Landscape of Modal Logics



- Stronger logics are higher in the diagram
- If a formula is **valid** in a weaker logic, it is **valid** in all stronger logics above it
- If a formula is **not valid** in a stronger logic, it is **not valid** in any weaker logic below it

Figure 1: Eight important modal logics

Example 4: $A \rightarrow \Box \Diamond A$ in KB

Example 4: $A \rightarrow \Box\Diamond A$ in KB

But this does hold in **KB**. Let's see this.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, A \rightarrow \Box \Diamond A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{F} 1.1, \Diamond A$ | $\Box \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1, A$ | $B \Diamond 4$ |
- ⊗

The $B \Diamond$ rule gives us the contradiction.

Example 5:

$$\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \text{ in } K$$

Example 5: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ in K

This is the K axiom - it should be valid even in K.

1. $\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ ✓ Assumption
2. $\mathbb{T} 1, \Box(A \rightarrow B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \rightarrow \Box B \rightarrow \mathbb{F} 1$

False conditional means true antecedent and false consequent.

- | | | |
|----|--|----------------------------|
| 1. | $\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \checkmark$ | Assumption |
| 2. | $\mathbb{T} 1, \Box(A \rightarrow B)$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box A \rightarrow \Box B \checkmark$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Box A$ | $\rightarrow \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1, \Box B$ | $\rightarrow \mathbb{F} 3$ |

The nested false conditional unpacks: $\Box A$ must be true and $\Box B$ must be false at world 1.

1.	$\models 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \checkmark$	Assumption
2.	$\top 1, \Box(A \rightarrow B)$	$\rightarrow \models 1$
3.	$\models 1, \Box A \rightarrow \Box B \checkmark$	$\rightarrow \models 1$
4.	$\top 1, \Box A$	$\rightarrow \models 3$
5.	$\models 1, \Box B \checkmark$	$\rightarrow \models 3$
6.	$\models 1.1, B$	$\Box \models 5$

False $\Box$ introduces world 1.1: B must be false there.

1.	$\models 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ ✓	Assumption
2.	$\top 1, \Box(A \rightarrow B)$	$\rightarrow \models 1$
3.	$\models 1, \Box A \rightarrow \Box B$ ✓	$\rightarrow \models 1$
4.	$\top 1, \Box A$	$\rightarrow \models 3$
5.	$\models 1, \Box B$ ✓	$\rightarrow \models 3$
6.	$\models 1.1, B$	$\Box \models 5$
7.	$\top 1.1, A \rightarrow B$	$\Box \top 2$
8.	$\top 1.1, A$	$\Box \top 4$

World 1.1 now exists, so both True $\Box$ rules fire: $A \rightarrow B$ is true at 1.1 (from line 2), and A is true at 1.1 (from line 4). Note these are reusable rules, so lines 2 and 4 stay unchecked.

1.	$\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box(A \rightarrow B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \rightarrow \Box B \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B \checkmark$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
7.	$\mathbb{T} 1.1, A \rightarrow B \checkmark$	$\Box \mathbb{T} 2$
8.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
$\swarrow \quad \searrow$		
9.	$\mathbb{F} 1.1, A \quad \mathbb{T} 1.1, B$	$\rightarrow \mathbb{T} 7$
	$\otimes \quad \otimes$	

The true conditional at 1.1 branches: either A is false at 1.1 (clashing with line 8) or B is true at 1.1 (clashing with line 6). Both branches close, so this is valid in $\mathbf{K}$.

Example 5: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ in K

Another way to think about this is that it's saying that if you have

- $\Box(A \rightarrow B)$; and
- $\Box A$

you can derive

- $\Box B$

The general principle is that if an argument is valid, then putting a box in front of every premise and in front of the conclusion will preserve validity.

Example 6:

$$(\Box A \vee \Box B) \rightarrow \Box(A \vee B) \text{ in } K$$

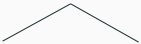
Example 6: $(\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ in K

1. $\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ ✓ Assumption
2. $\mathbb{T} 1, \Box A \vee \Box B \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box(A \vee B) \rightarrow \mathbb{F} 1$

False conditional: true antecedent and false consequent.

1. $\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ ✓ Assumption
2. $\top 1, \Box A \vee \Box B$ ✓ $\rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box(A \vee B)$ ✓ $\rightarrow \mathbb{F} 1$
4. $\top 1, \Box A \quad \top 1, \Box B$ $\vee \top 2$

The true disjunction branches: left branch has $\Box A$, right branch has $\Box B$.
Work on the left branch first.

1.	$\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box A \vee \Box B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(A \vee B) \checkmark$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{T} 1, \Box A \quad \mathbb{T} 1, \Box B$	$\vee \mathbb{T} 2$
5.	$\mathbb{F} 1.1, A \vee B$	$\Box \mathbb{F} 3$

False $\Box$ (line 3) introduces world 1.1 on the left branch: $A \vee B$ must be false there.

1.	$\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ ✓	Assumption
2.	$\top 1, \Box A \vee \Box B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(A \vee B)$ ✓	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\top 1, \Box A \quad \top 1, \Box B$	$\vee \top 2$
5.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Box \mathbb{F} 3$
6.	$\top 1.1, A$	$\Box \top 4$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 5$
	⊗	

World 1.1 exists, so True $\Box$ (line 4, $\Box A$) fires: A is true at 1.1. Then False $\vee$ gives $\mathbb{F} 1.1:A$, clashing with line 5. Left branch closes. Now the right branch.

1.	$\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box A \vee \Box B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(A \vee B)$ ✓	$\rightarrow \mathbb{F} 1$
$ \begin{array}{c} \swarrow \quad \searrow \\ \mathbb{T} 1, \Box A \quad \mathbb{T} 1, \Box B \end{array} $		
4.	$\mathbb{T} 1, \Box A$	$\vee \mathbb{T} 2$
5.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 5$
8.	$\otimes$ $\mathbb{F} 1.2, A \vee B$	$\Box \mathbb{F} 3$

False $\Box$ (line 3) introduces world 1.2 on the right branch: $A \vee B$ must be false there too.

1.	$\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box A \vee \Box B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(A \vee B)$ ✓	$\rightarrow \mathbb{F} 1$
$\swarrow \quad \searrow$		
4.	$\mathbb{T} 1, \Box A$ $\mathbb{T} 1, \Box B$	$\vee \mathbb{T} 2$
5.	$\mathbb{F} 1.1, A \vee B$ ✓	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 5$
8.	⊗ $\mathbb{F} 1.2, A \vee B$ ✓	$\Box \mathbb{F} 3$
9.		$\Box \mathbb{T} 4$
10.		$\vee \mathbb{F} 8$
11.		$\vee \mathbb{F} 8$
	⊗	

This is valid in K. Intuitively, if either one of $\Box A$ or $\Box B$ is true, we can derive $\Box(A \vee B)$.

Example 7:

$$(\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B) \text{ in } \mathbf{K}$$

Example 7: $(\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ in K

This should fail; different worlds might make $\Diamond A$ and $\Diamond B$ true.

And the special case where $B = \neg A$ also suggests it is invalid.

- | | | |
|----|--|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B) \checkmark$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A \wedge \Diamond B \checkmark$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond(A \wedge B)$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond A$ | $\wedge \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1, \Diamond B$ | $\wedge \mathbb{T} 2$ |

Start in the usual way; left-hand side true, right-hand side false. The left-hand side is a true $\wedge$, so I've just spelled it out.

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A \checkmark$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B \checkmark$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$

Lines 4 and 5 require introducing new worlds, and crucially we introduce different new worlds for each. The False $\Diamond$ rule (line 3) must be applied at each. Start with world 1.1.

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$ ✓	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 3$

$A \wedge B$ must be false at world 1.1. Now apply the same rule at world 1.2.

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$ ✓	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 3$
9.	$\mathbb{F} 1.2, A \wedge B$	$\Diamond \mathbb{F} 3$

Now we must apply the False $\wedge$ rule to lines 8 and 9. Start with line 8:
 $A \wedge B$ false at 1.1 branches into $\mathbb{F} 1.1:A$ or $\mathbb{F} 1.1:B$.

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B) \checkmark$	Assumption
2.	$\top 1, \Diamond A \wedge \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\top 1, \Diamond A \checkmark$	$\wedge \top 2$
5.	$\top 1, \Diamond B \checkmark$	$\wedge \top 2$
6.	$\top 1.1, A$	$\Diamond \top 4$
7.	$\top 1.2, B$	$\Diamond \top 5$
8.	$\mathbb{F} 1.1, A \wedge B \checkmark$	$\Diamond \mathbb{F} 3$
9.	$\mathbb{F} 1.2, A \wedge B$	$\Diamond \mathbb{F} 3$
$\swarrow \quad \searrow$		
10.	$\mathbb{F} 1.1, A \quad \mathbb{F} 1.1, B$	$\wedge \mathbb{F} 8$
	$\otimes$	

The left sub-branch closes: $\mathbb{F} 1.1:A$ clashes with $\top 1.1:A$ at line 6. The right sub-branch ($\mathbb{F} 1.1:B$) stays open. Now apply the False $\wedge$ rule to line 9.

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A \checkmark$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B \checkmark$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.1, A \wedge B \checkmark$	$\Diamond \mathbb{F} 3$
9.	$\mathbb{F} 1.2, A \wedge B \checkmark$	$\Diamond \mathbb{F} 3$
$ \begin{array}{c} \swarrow \quad \searrow \\ \mathbb{F} 1.1, A \quad \mathbb{F} 1.1, B \end{array} $		
10.	$\mathbb{F} 1.1, A$	$\wedge \mathbb{F} 8$
	$\otimes$	
$ \begin{array}{c} \swarrow \quad \searrow \\ \mathbb{F} 1.2, A \quad \mathbb{F} 1.2, B \end{array} $		
11.	$\mathbb{F} 1.2, A$	$\wedge \mathbb{F} 9$
	$\otimes$	

Example 8:

$$\Box(A \vee B) \rightarrow (\Box A \vee \Diamond B) \text{ in K}$$

Example 8: $\Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ in K

1. $\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ ✓ Assumption
2. $\mathbb{T} 1, \Box(A \vee B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \vee \Diamond B \rightarrow \mathbb{F} 1$

Assume the formula is false at world 1. A false conditional gives us a true antecedent ($\Box(A \vee B)$) and a false consequent ($\Box A \vee \Diamond B$).

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ ✓ | Assumption |
| 2. | $\top 1, \Box(A \vee B)$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box A \vee \Diamond B$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{F} 1, \Box A$ | $\vee \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1, \Diamond B$ | $\vee \mathbb{F} 3$ |

A false disjunction means both disjuncts are false: $\Box A$ is false and $\Diamond B$ is false at world 1. Note these are non-branching — we get both on the same branch.

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Box A$ ✓	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond B$	$\vee \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 4$
7.	$\mathbb{T} 1.1, A \vee B$	$\Box \mathbb{T} 2$
8.	$\mathbb{F} 1.1, B$	$\Diamond \mathbb{F} 5$

False $\Box$ (line 4) introduces world 1.1 with $\mathbb{F} 1.1:A$. Now the two reusable rules fire at world 1.1: True $\Box$ (line 2) gives $\mathbb{T} 1.1:A \vee B$, and False $\Diamond$ (line 5) gives $\mathbb{F} 1.1:B$. Lines 2 and 5 stay unchecked as reusable rules.

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ ✓	Assumption
2.	$\top 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \vee \Diamond B$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Box A$ ✓	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond B$	$\vee \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 4$
7.	$\top 1.1, A \vee B$ ✓	$\Box \top 2$
8.	$\mathbb{F} 1.1, B$	$\Diamond \mathbb{F} 5$
$\begin{array}{c} \swarrow \quad \searrow \\ \top 1.1, A \quad \top 1.1, B \end{array}$		
9.	$\top 1.1, A \quad \top 1.1, B$	$\vee \top 7$
	$\otimes \quad \otimes$	

The true disjunction at line 7 branches: either A is true at 1.1 (closing against $\mathbb{F} 1.1:A$ at line 6) or B is true at 1.1 (closing against $\mathbb{F} 1.1:B$ at line 8). Both branches close, so the formula is valid in $\mathbf{K}$.

Why is this valid?

Suppose $\Box(A \vee B)$ is true at world 1 but $\Box A \vee \Diamond B$ is false.

- Since $\Box A \vee \Diamond B$ is false, both $\Box A$ and $\Diamond B$ are false at world 1.
- Since $\Diamond B$ is false, B is false at all accessible worlds.
- So if at least one of A and B is true at each accessible world, as the premise says, and it's never B , it must always be A .
- But that would make $\Box A$ true, contradicting the assumption that it's false.

So the assumption that the formula is false leads to a contradiction. It must be valid in **K**.

Example 9: $\Diamond \Box A \rightarrow A$ in KT

Example 9: $\Diamond\Box A \rightarrow A$ in KT

1. $\mathbb{F} 1, \Diamond\Box A \rightarrow A$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond\Box A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, A \rightarrow \mathbb{F} 1$

Assume the formula is false at world 1. A false conditional gives a true antecedent ($\Diamond\Box A$) and a false consequent (A false at world 1).

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$ | $\Diamond \mathbb{T} 2$ |

True $\Diamond$ (line 2) introduces world 1.1: there must be an accessible world where $\Box A$ holds.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$ ✓ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1.1, A$ | $\mathbb{T} \Box 4$ |

The $\mathbb{T} \Box$ rule gives us A true at 1.1: since $\Box A$ is true at 1.1, and every world accesses itself in KT, A must hold at 1.1. But we have no way to get A true back at world 1, so the tree stays open. This formula is **not** valid in KT.

The open branch describes a model where the formula fails:

- There are two worlds: w_1 and w_2 , where w_1 accesses w_2 (and each world accesses itself, as required by the T condition).
- A is **false** at w_1 and **true** at w_2 .
- Since A is true at w_2 and w_2 only accesses itself, $\Box A$ is true at w_2 .
- Since w_1 accesses w_2 and $\Box A$ holds there, $\Diamond \Box A$ is true at w_1 .
- So $\Diamond \Box A$ is true but A is false at w_1 , and the formula fails.

The key point: $\Box A$ being true somewhere accessible to w_1 does not make A true at w_1 itself, unless accessibility is **symmetric**.